Worksheet 5.5 — Optimization

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

The Big Ideas

Optimization reduces a two-variable problem to one variable using a constraint, then finds a critical point.

- Use the constraint to eliminate a variable, differentiate, set to 0, confirm max/min.

Worked Examples

Example 1 — Maximize area with fixed fencing

A rectangle is enclosed by 160 m of fencing. Find the dimensions maximizing area.

Solution. $y = 80 - x$; $A = 80x - x^2$; $A' = 80 - 2x = 0 \Rightarrow x = 40$.

Conclusion: $x = y = 40$ m (a square).

Example 2 — Minimize fencing with fixed area

A pen must have area 484 m^2; find dimensions minimizing perimeter.

Solution. $y = 484/x$; $P = 2x + 968/x$; $P' = 2 - 968/x^2 = 0 \Rightarrow x^2 = 484 \Rightarrow x = 22$.

Conclusion: $x = y = 22$ m.

Example 3 — Open-top box

An open box is cut from a 18×18 cm sheet with corner squares of side x. Find x maximizing volume.

Solution. $V = x(18 - 2x)^2$; $V' = (18 - 2x)^2 + x \cdot 2(18 - 2x)(-2) = (18 - 2x)(18 - 6x)$; zero at $x = 9$ (rejected, collapses box) or $x = 3$.

Conclusion: $x = 3$ cm.

Example 4 — Minimum surface area cylinder

A cylinder must hold volume 432π cm^3; find the radius minimizing surface area.

Solution. $h = 432/r^2$; $S = 2\pi r^2 + 864\pi/r$; $S' = 4\pi r - 864\pi/r^2 = 0 \Rightarrow r^3 = 216$.

Conclusion: $r = 6$ cm.

Example 5 — Minimizing distance to a curve

Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$.

Solution. Minimize $D^2 = x^2 + (x^2 - 3)^2$ (using squared distance to avoid the root). $D^2 = x^4 - 5x^2 + 9$; derivative $4x^3 - 10x = 0 \Rightarrow 2x(2x^2 - 5) = 0$.

Conclusion: $x = 0$ or $x = \pm\sqrt{2.5}$; testing shows $x = \pm\sqrt{2.5}$ gives the minimum (closer than $x = 0$).

Example 6 — Maximizing a triangle's area with a fixed hypotenuse-style constraint

A rectangle is inscribed under the line $x + 2y = 20$ in the first quadrant (one corner at the origin, opposite corner on the line). Find the dimensions maximizing its area.

Solution. $y = \dfrac{20 - x}{2}$; $A = xy = \dfrac{x(20 - x)}{2} = 10x - \dfrac{x^2}{2}$; $A' = 10 - x = 0 \Rightarrow x = 10$.

Conclusion: $x = 10$, $y = 5$; max area $= 50$.

Example 7 — Minimizing material for an open cylinder (no lid)

An open-top cylindrical can (no lid) must hold volume 250π cm^3; find the radius minimizing material used.

Solution. $h = 250/r^2$; $S = \pi r^2 + 2\pi rh = \pi r^2 + 500\pi/r$ (only one circular base, since it's open-top); $S' = 2\pi r - 500\pi/r^2 = 0 \Rightarrow r^3 = 250$.

Conclusion: $r = \sqrt[3]{250} \approx 6.30$ cm.

Example 8 — Application — minimizing total travel cost

A delivery route's cost is $C(x) = 2x + \dfrac{288}{x}$ (\$100s) for average speed-related parameter $x > 0$. Find x minimizing cost.

Solution. $C'(x) = 2 - \dfrac{288}{x^2} = 0 \Rightarrow x^2 = 144$.

Conclusion: $x = 12$ minimizes cost (rejecting the negative root).

Example 9 — Confirming a maximum with the second derivative test

For the rectangle-under-a-line problem (Example 6), confirm $x = 10$ gives a maximum using A''.

Solution. $A(x) = 10x - \frac{x^2}{2} \Rightarrow A''(x) = -1 < 0$ for all x.

Conclusion: $A'' < 0$ confirms $x = 10$ is indeed a maximum.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

A rectangle is enclosed by 200 m of fencing; find dimensions maximizing area.

Question 2

A pen must have area 900 m^2; find dimensions minimizing perimeter.

Question 3 — Challenge

An open box is cut from a 20×20 cm sheet with corner squares of side x; find x maximizing volume.

Question 4

A cylinder must hold volume 2000π cm^3; find the radius minimizing surface area.

Question 5

For the cylinder above, find the height and confirm $h = 2r$ at the optimum.

Question 6 — Challenge

A farmer has 240 m of fencing for a rectangular field using an existing wall as one side (so only 3 sides need fencing); find the dimensions maximizing area.

Question 7 — Challenge

Find the point on the parabola $y = x^2$ closest to $(0, 2)$ (minimize squared distance).

Question 8

A rectangle is inscribed under the line $x + 3y = 30$ in the first quadrant; find the dimensions maximizing its area.

Question 9 — Challenge

An open-top cylindrical can (no lid) must hold volume 128π cm^3; find the radius minimizing material used.

Question 10

A delivery route's cost is $C(x) = 3x + \dfrac{432}{x}$; find x minimizing cost.

Question 11

For the inscribed-rectangle problem above ($x + 3y = 30$), confirm the maximum using the second derivative.

Question 12 — Challenge

A box with a square base and no lid must hold 108 cm^3; find the base side minimizing total material used.

Question 13 — Challenge

Two positive numbers have a sum of 24; find the two numbers that maximize their product.

Question Recap & Answer Key

Question Recap

1. A rectangle is enclosed by 200 m of fencing; find dimensions maximizing area.
2. A pen must have area 900 m^2; find dimensions minimizing perimeter.
3. An open box is cut from a 20×20 cm sheet with corner squares of side x; find x maximizing volume.
4. A cylinder must hold volume 2000π cm^3; find the radius minimizing surface area.
5. For the cylinder above, find the height and confirm $h = 2r$ at the optimum.
6. A farmer has 240 m of fencing for a rectangular field using an existing wall as one side (so only 3 sides need fencing); find the dimensions maximizing area.
7. Find the point on the parabola $y = x^2$ closest to $(0, 2)$ (minimize squared distance).
8. A rectangle is inscribed under the line $x + 3y = 30$ in the first quadrant; find the dimensions maximizing its area.
9. An open-top cylindrical can (no lid) must hold volume 128π cm^3; find the radius minimizing material used.
10. A delivery route's cost is $C(x) = 3x + \dfrac{432}{x}$; find x minimizing cost.
11. For the inscribed-rectangle problem above ($x + 3y = 30$), confirm the maximum using the second derivative.
12. A box with a square base and no lid must hold 108 cm^3; find the base side minimizing total material used.
13. Two positive numbers have a sum of 24; find the two numbers that maximize their product.

✓ Answer Key

1. $y = 100 - x$, $A' = 100 - 2x = 0$: $x = y = 50$ m
2. $y = 900/x$, $x^2 = 900$: $x = y = 30$ m
3. $V = x(20 - 2x)^2$; $V' = (20 - 2x)(20 - 6x)$: $x = 10$ (rejected) or $x = \frac{10}{3}$ cm
4. $S = 2\pi r^2 + 4000\pi/r$; $S' = 4\pi r - 4000\pi/r^2 = 0 \Rightarrow r^3 = 1000 \Rightarrow r = 10$ cm
5. $h = 2000/100 = 20$; $2r = 20 = h$ confirmed
6. Let $x =$ side perpendicular to wall, $240 - 2x =$ side parallel; $A = x(240 - 2x) = 240x - 2x^2$; $A' = 240 - 4x = 0 \Rightarrow x = 60$, parallel side $= 120$ m
7. minimize $D^2 = x^2 + (x^2 - 2)^2 = x^4 - 3x^2 + 4$; $4x^3 - 6x = 0 \Rightarrow 2x(2x^2 - 3) = 0 \Rightarrow x = \pm\sqrt{1.5}$; point $(\sqrt{1.5}, 1.5)$ (or its negative-x mirror)
8. $y = \frac{30-x}{3}$; $A = xy = 10x - \frac{x^2}{3}$; $A' = 10 - \frac{2x}{3} = 0 \Rightarrow x = 15$, $y = 5$
9. $S = \pi r^2 + 256\pi/r$; $S' = 2\pi r - 256\pi/r^2 = 0 \Rightarrow r^3 = 128 \Rightarrow r = \sqrt[3]{128} \approx 5.04$ cm
10. $C'(x) = 3 - 432/x^2 = 0 \Rightarrow x^2 = 144 \Rightarrow x = 12$
11. $A''(x) = -\frac{2}{3} < 0$ for all x, confirming a maximum
12. $S = x^2 + 4x(108/x^2) = x^2 + 432/x$; $S' = 2x - 432/x^2 = 0 \Rightarrow x^3 = 216 \Rightarrow x = 6$ cm
13. let numbers be $x, 24 - x$; $P = x(24 - x) = 24x - x^2$; $P' = 24 - 2x = 0 \Rightarrow x = 12$: both numbers are 12